

Future research directions

Options after “Driving in the wrong direction? Modelling policy mixes for EV adoption”

Abstract

Fourteen candidate directions for extending the agent-based model of the ICEV–EV transition, grouped into five themes, each assessed against three questions: which limitation of the published paper does it address, what already exists in the repository, and what does it cost — in particular, whether it forces a re-run of the simulation-based calibration. The concluding sections give a summary table and a recommended sequence. The central practical finding is that the directions differ in cost by more than an order of magnitude, and that the two cheapest are also the two that lift limitations the published paper states explicitly.

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1 Purpose and how to read this

The published model is calibrated on California 2001–2023 and used to evaluate five market-based instruments, individually and in pairs, over 2024–2035 with a projection to 2050. It is a working, calibrated, documented artefact. Every direction below is therefore an *extension*, not a rebuild, and the relevant question for each is not “is this interesting” — they all are — but “what does it cost, what does it buy, and does it break the calibration”.

Three cross-cutting facts govern everything that follows.

The calibration is the bottleneck. Only four parameters are estimated by simulation-based inference: a_χ , b_χ (the innovativeness-threshold Beta distribution), κ (consumer intensity of choice) and δ (fuel-efficiency depreciation). See `package/calibration/NN_multi_round_calibration_multi_gen.py`. Any extension that changes agent behaviour *during* 2001–2023 invalidates that posterior and forces a re-run. Extensions confined to the 2024–2035 policy period do not. This single distinction separates the cheap directions from the expensive ones and is the axis the summary table in Section 10 is sorted on.

No firm behavioural parameter is estimated. λ (research intensity of choice, 10^{-3}), M^+ (memory, 30), Y^+ (variant cap, 10), θ and ι (activation probabilities) are all set by assumption, as Table 4 of the manuscript records. The firm side is disciplined only indirectly, through the price, HHI and car-age target ranges of Table 1. This matters because it means the supply-side extensions (Themes A, C, D) are not competing with estimated parameters — they are replacing one asserted specification with another, and the case for each has to be made on economic grounds rather than fit.

Several directions are already partly built. Forward-looking expectations and three command-and-control bans are implemented in the current model code behind default-off flags. A Gaussian-process surrogate over the full five-instrument policy space, and a ZIP-level spatial calibration pipeline with a 28-dimensional summary statistic, both exist in git history and were removed from the publication snapshot. Each section below records this under `STATUS:` .

2 What the paper commits us to

The conclusions section of the manuscript names five future directions of its own. Mapping them onto the numbered ideas is the cheapest available source of referee-proof motivation, because a limitation the paper itself raises does not need to be argued for.

Stated in the paper	Corresponding direction
“A key shortcoming ... is the lack of financial and physical constraints ... Future research should incorporate capital costs of switching production and a supply model with an explicit representation of production planning and inventory management.”	Items 5 (financial constraints), 6 (switching and R&D costs), 12 (batch/vintage production). Theme C.
“Incorporating more sophisticated forecasting techniques and factoring in expected regulation can crucially modify investment and purchase decisions. By doing so, policies such as standards and regulations could be assessed. Additionally, this would enable the study of more realistic time-varying carbon price implementations such as the EU-ETS.”	Items 1 (forward-looking agents), 2 (command and control), 4 (long-horizon firm profit), 7 (MARL firms). Theme A. Also the time-varying half of item 8.
“The role of pro-environmental attitudes or conspicuous consumption ... could be better represented by expanding ... beyond a threshold model to enrich car users with social norms and behavioural control such as in the Theory of Planned Behaviour ... This would facilitate the inclusion of policies such as information provision, media or marketing.”	Not on the current list. See Section 12.
“[T]ranslating these results into concrete recommendations would benefit from moving beyond calibration to validation. Achieving this requires a larger car user population to reduce noise ... enabling time-series data on EV fleet size or sales to be split into training, validation, and test sets.”	Items 9 (spatial), 10 (population synthesis). Theme E.
“[W]e restrict our analysis to policy pairs rather than exhaustive combinations to maintain tractability” (Section 2) and policies “are applied from January 2024 at their full intensity and are then constant over the policy period” (Section 4.1).	Item 8 (multi-policy optimisation). Theme B.

3 The five themes

Theme A — Expectations, anticipation and strategic firms. Items 1, 2, 4, 7. All four attack the same sentence in Section 4.1 of the manuscript: “we disregard the role of expectations regarding policy stringency ... with agents assuming that policies will be permanent.” They form a ladder of increasing cost and decreasing certainty, and the rungs are not independent: item 2 is only meaningful given item 1, and item 7 delivers little that items 1 and 4 do not deliver more cheaply.

Theme B — Policy-space search and the planner’s problem. Item 8. Does not change the model at all; changes how the policy space is searched. The only theme with no recalibration exposure.

Theme C — Supply-side economics: constraints, costs, capacity. Items 5, 6, 12. The firm side is where the model is most stylised, and this is the paper’s own first-named limitation.

Theme D — Technology representation. Items 3, 13, 14. How the NK landscape, battery technology and production carbon intensity are specified. Infrastructure work with one exception (item 14) that touches a headline result cheaply.

Theme E — Spatial grounding and empirical identification. Items 9, 10, 11. The most expensive theme and the one with the strongest claim on the model’s scientific credibility, because it is the only theme that can defend the paper’s *core mechanism* against its most obvious critique.

4 Theme A: expectations, anticipation and strategic firms

4.1 Item 1 — Forward-looking agents

STATUS: implemented, documented, default off. See `docs/forward_looking/forward_looking_expectations.tex` for the full derivation. The flag `forward_looking_expectations` replaces the naive closed-form geometric series

$$\text{LCC}_{\text{extra}} = D_i \frac{(1+r)(1-\delta)(c+\gamma_i e)}{\omega(r-\delta-r\delta)} \quad (1)$$

— which assumes today’s fuel cost and emissions intensity persist forever — with a backward recursion $\text{Index}(t) = z_t + q \text{Index}(t+1)$ over the model’s actual known future path. It is applied consistently to consumers (`socialNetworkUsers.py`) and firms (`firm.py:_lifecycle_cost_term_firm`), and the carbon-price component is extended past the simulated horizon using `calculate_price_at_time()` so that a forward-looking agent correctly believes a temporary carbon tax is temporary rather than permanent at its peak.

Why it matters. The manuscript’s own Section 2.2 flags the assumption — “car users assume the future prices and emissions of gasoline and electricity to be the same as in the present period” — and justifies it by citing Anderson et al. (2013). That justification is reasonable for *gasoline price* expectations, which is what Anderson et al. study. It is much weaker for a pre-announced policy ramp, which is public information rather than a forecasting problem.

Effect size is already known to be large. The `car_ban` smoke test recorded in the forward-looking note found that because $q = 1/((1+r)(1-\delta)) \approx 0.9977$, the effective planning horizon is $1/(1-q) \approx 433$ months. A driving ban six to eleven years out is therefore already 70–85% priced in from the start of the future period. Forward-looking EV fleet share reached 0.97 six months *before* a 2030 ban took effect, against 0.39 for naive agents facing the identical ban. Anticipation is not a marginal correction here; it is the dominant channel.

Cost. One SBI re-run. The flag changes calibration-period behaviour too, because gasoline prices, electricity prices and grid decarbonisation all vary over 2001–2023 even though the carbon price is forced to zero there. It also changes how the calibrated δ enters utility, so δ in particular should be expected to move.

4.2 Item 2 — Command-and-control policies

STATUS: three bans implemented in `controller.py`; experiment packages removed from the snapshot. `ICE_driving_ban_time` / `ICE_driving_ban_penalty` (a cost shock fed through the same discounting machinery as item 1, so anticipation emerges from the existing utility function with no separate “anticipation lead” parameter), `ICE_sales_ban_time` (a hard constraint on firms’ choice sets, which also flushes pre-ban ICE stock from `cars_on_sale`) and `ICE_research_ban_time` (no ICE neighbouring technologies generated at all) all exist and all default to `None`. The driving-ban experiment lived in `package/car_ban` at commit 366d91a and is recoverable from git.

Why it matters most for this paper specifically. California is the case study *and* the paradigmatic command-and-control jurisdiction — the manuscript says so in Section 3, cites the ZEV mandate, and then explicitly declines to model it, on the grounds that “regulatory measures are out of the scope of the present research”. That is a defensible scoping decision for one paper and an obvious opening for the next. A paper that puts the ZEV mandate back in, and compares it against the market-based instruments already characterised, has a ready-made motivation, a ready-made baseline, and an audience that is already asking the question.

Cost. Essentially free *given item 1*. Bans are configured as absolute timesteps after the burn-in and can be confined to the future period, so they do not touch the calibration. But a ban that agents cannot anticipate is just a cost shock arriving on a known date, which is not what a ZEV mandate is. Item 2 without item 1 answers the wrong question.

4.3 Item 4 — Firms chasing long-horizon profit

STATUS: not implemented. The cheapest unbuilt item on the list.

The manuscript’s Section 2.3 says manufacturers “select the car designs to put on sale based on their profit expectations throughout the lifecycle of the technology”. The implementation is narrower than that: Equation 8’s expected profit, as computed in `firm.py:calc_predicted_profit_segments_research`, is a *single-period* quantity — marginal profit $\times$ segment population $\times$ expected in-segment share, all evaluated at this month’s prices, this month’s subsidy scalar and this month’s segment sizes. A firm choosing a research direction today receives a technology that will sell for years, and scores it on one month of profit.

The fix needs no learning. Replace the one-period profit in Equation 12’s softmax with a discounted profit stream over the product’s expected commercial lifetime, using the discount rate r the model already has. Two sub-channels are worth separating because they can be implemented and reported independently:

1. **Policy instruments on the firm’s own P&L.** The production subsidy and the new-EV rebate enter `firm.next_step()` as scalars. A firm weighing an R&D direction should weight them over the payoff window, not this month. This is the same backward recursion item 1 already implements, applied to the subsidy vectors.

2. **Segment growth.** $|I_{s,t}|$ and $W_{s,t}$ are today’s segment sizes and competitive state. But EV consideration spreads endogenously as χ thresholds are crossed — which is precisely the mechanism policy is trying to trigger — and firms are blind to it. This sub-channel needs a model of segment evolution, however crude, and is where the honest ceiling on a non-learning implementation sits.

A predicted tension worth stating in advance. Two opposing forces act here. Forward-looking firms that see the carbon ramp coming should pivot to EVs *earlier*, lowering the policy intensity needed to hit 95%. But Figure S6 panel G reports that EV uptake *increases* when research choices are more random, because policy incentives favouring EVs outweigh the ICE research advantage accumulated during burn-in — so making firms sharper should push the other way. Against that, the Sobol indices (Figures S7, S8) attribute almost no global variance in EV adoption to λ . The reasonable reading is that the *level* of firm rationality is not first-order in the current specification, but whether firms anticipate *policy* may well be. That asymmetry is an argument for prioritising items 1 and 4 over item 7.

4.4 Item 7 — Strategic, pre-emptive firms via MARL

STATUS: not implemented. Highest risk on the list.

Replace the myopic softmax over NK neighbours with a learned policy $\pi(a \mid s)$ trained to maximise discounted profit. Four objections, in descending order of seriousness:

1. **It breaks the calibration chain.** SBI draws θ and re-simulates thousands of times. If each draw needs trained firms, the nested loop is infeasible. There are two ways out: freeze one policy trained at the posterior mode and hold it across draws (defensible, since λ is held fixed across draws today), or *amortise* — put θ in the firm’s observation vector and sample it from the SBI prior during training, so one trained policy is valid at any θ . The latter is the technically right answer and has a pleasing symmetry with what NPE already does: train once over the prior, valid everywhere.
2. **It trades one interpretable knob for several uninterpretable ones.** λ has an economic reading (exploration versus exploitation). Learning rate, network width, training length and exploration schedule do not, and they move outcomes. Any result has to be shown invariant to them, or reported as a range. Budget for this; it is the part referees will find.
3. **Multi-agent learning is non-stationary.** Ten co-learning firms have no convergence guarantee, and outcomes become sensitive to algorithmic detail — a poor property for a model whose output is a policy recommendation. Best-response iteration (fix firms, train planner; fix planner, train firms; repeat and check for a fixed point) is far more stable and more reportable than simultaneous learning.
4. **Most of its value arrives more cheaply via items 1 and 4.** Against a *pre-announced deterministic* policy path — which is what `calculate_price_at_time()` produces — anticipation requires no learning at all. A firm can read the schedule. MARL becomes genuinely necessary only once the planner itself is adaptive (the RL half of item 8), because then the policy path depends on the state, which depends on what firms do, and no closed form exists.

If pursued, keep it minimal. Do not learn pricing — `calc_optimal_price_cars` is a closed-form optimum (Equation 10, via the Lambert W function) and learning can only degrade it. Replace only `select_car_lambda_research`, or even just let the agent set a scalar EV/ICE direction bias

that shifts the profit scores. Share one policy network across all ten firms, which avoids ten-way non-stationarity and is defensible as an industry-standard heuristic.

The result that would justify it. An adaptive regulator that visibly raises subsidies when uptake stalls gives firms an incentive to slow-roll EV investment. If the flexibility gain from an adaptive planner partly evaporates once firms learn to exploit it, that is a commitment-versus-discretion finding, and it is a real contribution rather than a methods demonstration. But it presupposes item 8.

5 Theme B: policy-space search and the planner’s problem

5.1 Item 8 — Multi-policy optimisation

STATUS: Gaussian-process surrogate fully prototyped at commit c92bff8; removed from the publication snapshot.

The recoverable `package/surrogate/` contained:

- `sampling.py` — Latin hypercube design over the full five-dimensional policy space, evaluating four outputs (`ev_uptake`, `utility`, `emissions`, `net_cost`).
- `surrogate.py` — a Gaussian process with a Matérn 5/2 kernel plus a `WhiteKernel` nugget to absorb seed-to-seed variance, with leave-one-out cross-validation and prediction-interval coverage diagnostics. (The term in the original idea list, “Gaussian mixture model”, is worth correcting: a GMN/mixture density network is what `sbi` uses for the *posterior*; what is wanted here is a Gaussian *process* emulator of the policy response surface. Both already exist in this project, doing different jobs.)
- `optimisation.py` — active Bayesian optimisation with a constrained acquisition function, $EI(x) \times \Pr(\text{EV uptake} \in [\underline{u}, \bar{u}] \mid x)$, which reduces to pure feasibility-seeking before the first feasible point is found; then Pareto-front extraction over ~ 170 evaluated policies via randomly weighted scalarisations.

Why this is the strongest near-term option. It lifts two limitations that the paper states are computational rather than scientific. First, pairs: “we restrict our analysis to policy pairs rather than exhaustive combinations to maintain tractability”. The paper’s own finding of perfect substitutability between adoption and production subsidies (Section 4.2.2, the horizontal line in Figure 4) is a two-dimensional slice of a surface a surrogate can map properly. Second, constant intensities: policies are “applied from January 2024 at their full intensity and are then constant”. A surrogate over instrument *and ramp shape* answers the EU-ETS question the conclusions raise.

And it is cheap. No model change, therefore no recalibration, therefore no new behavioural assumption to defend. It is the only direction on this list with essentially zero scientific risk: if the GP fits badly the diagnostics say so, and the fallback is more design points.

The RL-planner variant. An alternative to a surrogate over fixed schedules is a reinforcement-learning planner emitting $\pi(a_t \mid s_t)$ monthly. This is a different and larger claim — it answers “how much cheaper is 95% uptake if the regulator can *adapt*?” rather than “what is the best fixed mix?”. It is attractive because the model is already shaped like an RL environment: `single_`

`policy_simulation` loads a pickled post-calibration controller and runs only the 144-step future period, so the 64 saved controllers are ready-made episode resets with built-in seed diversity, and `calc_net_policy_distortion()` is ready-made reward. The main engineering cost is that `manage_policies()` precomputes policy vectors read by index, so per-step action injection means turning those reads into a setter. Two design cautions: use common random numbers (fix the controller batch per update) or gradient variance across 64 stochastic seeds will dominate; and restrict the observation vector to what a regulator actually observes (EV share, sales, prices, HHI, fuel prices), not firm internals, or the learned rule is not implementable and reviewers will say so.

Recommendation within item 8. Do the surrogate first — it is built, it is the direct answer to a stated limitation, and it produces the baseline the RL planner would have to beat.

6 Theme C: supply-side economics

6.1 Item 5 — Financial constraints on firms, budgets for consumers

STATUS: not implemented. The paper’s first-named limitation, and it is two separable ideas.

Firm side. Section 2.3 states the assumption plainly: firms “face constant marginal production costs and are not subject to binding capacity or financial constraints”, defended on the grounds that the relevant manufacturers are multinationals with capital-market access. The defence is reasonable and the paper cites Mazzucato and Unruh in acknowledging what it gives up. Note what the current specification implies: cumulative firm profit is a *reported output* with no feedback into innovation capacity. Adding a financing channel closes that loop, which is attractive because the co-evolutionary feedback *is* the paper’s headline methodological contribution.

Consumer side, which I think is the more interesting half. β_i scales with the Californian income distribution, but there is no budget constraint — a low-income agent can select an arbitrarily expensive car if the utility arithmetic works out. Figure 6 shows the β_i distribution has an extremely long right tail (to $\sim 7 \times 10^6$). A hard budget or a financing constraint (monthly payment as a share of income, which is how car purchases are actually constrained) would bite hardest exactly where the paper’s used-car-market contribution lives, and would sharpen the distributional reading of the rebate results. Given that Figure 7 shows removing the used-car market roughly *triples* EV uptake by forcing price-sensitive users out of cheap ICEVs, the interaction between budgets and the used market is likely to be first-order.

Cost. High. Both halves change calibration-period behaviour, so recalibration is required, and the consumer-budget version probably requires re-deriving the analytic β median (Supplementary Equation 5), which is currently pinned by matching the mean observed new-car price.

6.2 Item 6 — Switching costs and R&D costs

STATUS: not implemented. The most surgical item in this theme.

R&D is currently free: firms are gated only by the activation probability ι and by NK locality. Production switching is likewise free, gated only by θ . So the sole source of inertia in the supply side is technological path dependence through the landscape — which is exactly the “carbon lock-in” that Section 2.3 cites Unruh (2000, 2002) for and then argues is not binding.

Why it earns its place. A per-attempt R&D cost and a capital cost of switching a production line give the production subsidy and the profit channel real teeth, and they give accumulated profit a *use*. It is also the cheapest way to test whether the paper’s transition speeds are an artefact of costless pivoting. My prior is that they partly are, and that adding these costs raises the required policy intensities — pushing in the same direction as item 4’s rationality channel and the opposite direction to item 1’s anticipation channel. The three together would make a coherent robustness paper on whether the Table 3 intensities are conservative or optimistic.

Cost. Moderate. Recalibration required, and one or two new parameters that will have to be set by assumption plus sensitivity analysis, in the manner Section 3 already establishes for λ , K and μ .

6.3 Item 12 — Stock, vintage or batch production

STATUS: not implemented.

Production is on demand: Appendix A.1.1 is explicit that a new design $y \in Y_t$ “is represented by a vector of attributes and a sale price, but does not represent an actual car”, and a car is produced only when bought. So there is no inventory, no capacity constraint and no lead time.

It is exposed to a headline result. The paper’s second key takeaway is that “before 2030, all transition policies entail drawbacks. The required large-scale deployment of EVs causes an increase in manufacturing emissions that exceeds the savings in gasoline use.” Production emissions therefore track sales exactly, month by month, with no smoothing. Batch or vintage production with inventory would change the timing and possibly the magnitude of that manufacturing-emissions spike — which is the mechanism behind one of the paper’s two main policy warnings. That makes this less of an “add realism” item and more of a robustness check on an existing claim, which is a better framing for it.

Cost. High, and structural. This is the largest single change to the model’s architecture on the list.

7 Theme D: technology representation

7.1 Item 3 — Rethinking the NK landscape

STATUS: not implemented. Important, but poorly shaped as a paper.

Two concrete, already-documented problems:

1. **Range compression.** Because an attribute’s performance is the mean of $N = 15$ i.i.d. draws, normalised performance concentrates in $(0.2, 0.8)$ with mean and median 0.5 (Supplementary Section 3.2). The landscape bounds (F^-, F^+) therefore have to be back-solved analytically from the target attribute range (Q^-, Q^+) via Supplementary Equations 22–28 — machinery that exists purely to compensate for the concentration. The reachable attribute space is narrower than the nominal bounds suggest, and gets narrower as N grows, which is in tension with wanting N large enough for normality.
2. **K is confounded with the landscape seed.** The supplementary says so itself, at the end of Section 2.1: “by varying the complexity of the landscape, we change the inputs. This means

that differences between complexity runs may also be due to different seeds in the landscape generation.” So the reported K sensitivity is not clean, and K_{ICE} is described as having a “mixed but significant effect” on EV uptake.

A third, smaller symptom: simulated ICEV driving range is $\sim 20\%$ too high (Supplementary Section 1.4), attributed to pinning the fuel tank at the top tier of observed data.

My view. Fix the range compression and the K -seed confounding as *infrastructure*, folded into whichever substantive paper comes next, and report it in that paper’s supplementary. Do not try to make it the paper. There is a publishable version — “does the innovation representation drive the policy conclusion?”, comparing NK against an alternative such as a continuous cumulative-innovation or learning-curve formulation — but it is a methods-venue paper with no policy question attached, and it would consume a calibration cycle to produce a null result if the answer is “no”.

7.2 Item 13 — EV battery technology evolution

STATUS: partly present as an NK dimension. Battery capacity is already the extra fourth EV attribute, bounded (0,150) kWh from Schmuch et al. (2018), with $\rho(C,B) = 0$ so that cheap high-capacity EVs are hard to reach. So this item is largely a *special case of item 3*: making battery evolution richer means either changing the landscape specification or bolting an exogenous technology trend onto it. Worth doing as part of item 3 rather than as a separate direction; the one genuinely distinct sub-idea is an endogenous learning curve (cost falling in cumulative EV production), which is a different mechanism from NK search and would interact with items 6 and 12.

7.3 Item 14 — Time-varying carbon intensity of car production

STATUS: not implemented. Best cost-to-payoff ratio on the list.

Manufacturing emissions are *fixed constants* for the whole 2001–2050 horizon: ICEV 10 tonnes, EV 14 tonnes (Table 4, from IPCC 2022). Battery manufacturing has decarbonised substantially over exactly the period the model spans, and the model already assumes 90% grid decarbonisation by 2035 for *driving* emissions while holding *production* emissions flat — an internal inconsistency that is easy to state and easy to fix.

Why it matters more than it looks. It bears directly on two results. The pre-2030 manufacturing-emissions penalty (Section 4.2.1, and the third row of Figure 3) is computed with a constant 14 t per EV. And Supplementary Section 2.4 finds grid decarbonisation matters little for *uptake* but does matter for emissions *flow*. A declining production intensity would mechanically improve the EV case and could soften or shift the “all policies entail drawbacks before 2030” conclusion.

Cost. Low. It is an exogenous time series, structurally identical to the grid-decarbonisation and fuel-price paths `manage_scenario()` already handles. It does change LCE during 2001–2023, so it does touch the calibration — which is an argument for batching it with item 1 rather than doing it alone.

8 Theme E: spatial grounding and empirical identification

8.1 Items 9 and 10 — Spatial data and population synthesis

STATUS: far more built than the idea list suggests. Commit `e7881a9` (“setting up the code for the zip level synthetic population data set”) contains a complete, tested-against-placeholder-data pipeline:

- `package/model/` — `synthetic_population.py`
- `package/calibration/` — `summary_stats.py`, `calibration_data_outputs_zip.py`, `NN_multi_round_calibration_zip_gen.py`
- `package/constants/` — `base_params>NN_zip.json`
- `package/generating_data/` — `fake_zip_data_gen.py`, plus a documented column contract (`FAKE_DATA_README.md`) specifying exactly what real files must supply

The placeholder population cache (61,979 rows) is still in the current tree under `package/calibration_data/FAKE_zip_data/`, carrying `zip`, `weight`, `income`, `vmt`, `age`, `political_leaning`, `ruralness`, `latitude` and `longitude`.

The designed summary-statistic vector is 28-dimensional:

- 8 state EV stock shares (2016–2023) — the existing aggregate anchor;
- 8 state EV sales shares — already loaded by the current calibration and then never used;
- 9 cross-sectional gradients: weighted OLS of ZIP EV share on z-scored log income, ruralness and political leaning, at three years;
- 3 between-stratum dispersions of EV share;
- 3 Moran’s I values of ZIP EV share, on a k -nearest-neighbour weight matrix built from ZIP centroids.

The identification argument, which is the real reason to do this. a_χ is by far the dominant parameter for EV adoption — first-order Sobol index ≈ 0.78 (Figure S7) — and it governs the paper’s headline mechanism, the social-imitation threshold. But it is currently identified from an *aggregate time series* of EV uptake. Aggregate uptake cannot distinguish social contagion from preference sorting: both produce S-curves. That is the most obvious available attack on the published paper, and this pipeline is the answer to it. The design note in `summary_stats.py` states the logic precisely: pure sorting produces a spatial pattern whose shape is fixed from $t = 0$ and merely scales up, while contagion produces spatial autocorrelation that *grows* as adoption propagates outward — so it is the *trajectory* of Moran’s I across three separated years, not its level, that carries the identifying information.

It also delivers the paper’s own stated route to validation, because ZIP-level panel data supports genuine training/validation/test splits in a way one aggregate series does not.

The real blocker is data acquisition, not method. What is needed is ZIP-level EV registration stock and sales for California (CEC / DMV publish at this granularity), ACS income and age by ZCTA, a Census ZCTA gazetteer for centroids, and a ruralness and political-leaning index. This is legwork rather than research risk — which makes it a very different kind of cost from items 5, 7 or 12.

The one genuine technical blocker. At $I = 3000$ across ~ 250 ZIPs, there are roughly 12 agents per ZIP, which is far too noisy for per-ZIP moments. The gradient design partly works around this by pooling across ZIPs in a weighted regression rather than binning, but the population still has to grow substantially — which is precisely the “larger car user population to reduce noise . . . through model parallelisation” that the conclusions call for. Expect this to be the dominant engineering cost, and note it compounds with everything else: a larger population makes every recalibration in every other theme more expensive.

8.2 Item 11 — Charging infrastructure

STATUS: not implemented; deliberately scoped out.

Section 2.2 states that “the charging infrastructure . . . is a given in this model”, and Section 3 frames the whole contribution as “a transferable framework for evaluating market-based instruments in contexts where a sufficient charging infrastructure threshold has been reached”, citing Springel (2021) for the claim that infrastructure has limited additional effect past a threshold.

This is a stronger scoping argument than the others. Unlike the expectations assumption or the constant-production-emissions assumption, this one comes with a cited empirical justification and an explicit statement of the domain of validity. Adding infrastructure therefore does not fix an acknowledged weakness so much as *extend the model to a new domain* — jurisdictions below the infrastructure threshold, which is most of the world and arguably the more policy-relevant case. That is a legitimate and attractive project, but it should be framed as a domain extension rather than a correction, and it is only tractable after item 9 because charging supply is intrinsically spatial. Treat it as strictly downstream of Theme E.

9 The recalibration bottleneck

The single most useful piece of project planning here is to notice which items share a calibration cycle. An SBI re-run is the expensive, serialising step, so items that require one should be *batched into a single recalibration* rather than done sequentially.

No recalibration: item 2 (bans are future-period only), item 8 (operates on saved post-calibration controllers).

Recalibration required: items 1, 3, 4, 5, 6, 7, 12, 13, 14 (all change 2001–2023 behaviour), and items 9, 10, 11 (which redefine the calibration target itself).

Note that the two directions requiring no recalibration are exactly the two that address explicitly stated limitations. That is not a coincidence: the paper scoped out what was expensive.

A corollary for Theme E: because items 9 and 10 *change the calibration target* rather than merely invalidating the posterior, they should come either first — before behavioural changes accumulate — or last, after they have all settled. Interleaving a spatial recalibration with behavioural extensions means paying for both at once and being unable to attribute a change in fit to either.

10 Summary

Item	Built?	Recal.	Cost	What it buys
8 Multi-policy optimisation	mostly	no	low	Full 5-instrument Pareto front; time-varying schedules. Lifts a stated limitation.
2 Command and control	mostly	no	low	ZEV mandate on the ZEV state. Needs item 1.
1 Forward-looking agents	yes	yes	low	Removes the permanent-policy assumption. Effect known to be large.
14 Production carbon intensity	no	yes	low	Fixes an internal inconsistency; touches the pre-2030 emissions result.
4 Long-horizon firm profit	no	yes	low-mod	Aligns the code with what Section 2.3 already claims. Non-RL route to most of item 7.
6 Switching / R&D costs	no	yes	mod	Tests whether transition speeds are an artefact of costless pivoting.
3 NK landscape	no	yes	mod	Fixes range compression and K -seed confounding. Infrastructure, not a paper.
13 Battery evolution	partly	yes	mod	Largely a special case of item 3, except the learning-curve variant.
9 + 10 Spatial / pop. synthesis	mostly	target	high	Identifies contagion vs. sorting. Defends the core mechanism. Blocker is data, not method.
5 Financial / budget constraints	no	yes	high	Paper’s first-named limitation. Consumer-budget half interacts with the used-car contribution.
12 Batch / vintage production	no	yes	high	Robustness on the manufacturing-emissions spike. Largest architectural change.
11 Charging infrastructure	no	yes	high	Extends the model below the infrastructure threshold. Downstream of item 9.

Item	Built?	Recal.	Cost	What it buys
7 MARL firms	no	yes	very high	Commitment vs. discretion. Presupposes item 8; breaks the calibration chain.

11 Recommended sequence

First, and immediately: item 8 (surrogate). It is built, it needs no recalibration, it introduces no behavioural assumption to defend, and it answers a question the paper explicitly says it could not afford to answer. Restore `package/surrogate/` from `c92bff8`, re-point it at the current `base_params_endogenous_policy_pair_gen.json`, validate the GP (R^2 , leave-one-out, interval coverage), and extend the design space from five intensities to five intensities plus ramp shape. Lowest-risk paper available.

Second: items 1 + 2 + 4 as one bundle — “anticipation and regulation”. One recalibration buys all three. Item 1 is already implemented; item 2’s three bans are already implemented; item 4 is a contained change to one profit calculation. The motivation is quoted verbatim from the paper’s own conclusions, the case study is the jurisdiction that actually has a ZEV mandate, and the driving-ban smoke test already indicates the effect is large rather than marginal. Fold item 14 into the same recalibration since it is nearly free once you are paying for one.

Third, as the long-term bet: Theme E (items 9 + 10). This is the one that protects the published paper. a_χ carries the headline mechanism and dominates the Sobol decomposition, and it is currently identified from an aggregate series that cannot separate contagion from sorting. The pipeline for fixing that is designed, written and tested against placeholder data; what is missing is real ZIP-level California data and a larger population. Start the data acquisition now, in parallel with the first two projects, because it is the long-lead item and it is administrative rather than intellectual work.

Defer: items 7, 12, 5. Each is defensible and each is expensive. Item 7 in particular should wait until item 8’s RL-planner variant exists, because without an adaptive planner there is nothing for learning firms to learn that reading the announced schedule would not deliver. Items 5 and 12 are natural if a collaborator wants the industrial -organisation angle; they are not the best use of a solo calibration cycle.

Fold in rather than pursue: items 3, 13. Fix the NK range compression and the K -seed confounding inside whichever project comes next and report it in that supplementary.

A caution on breadth. Fourteen directions is a portfolio, not a plan, and they point in genuinely different directions: Theme B is computational, Theme A is behavioural, Theme C is industrial organisation, Theme E is econometric. Three of these done properly is a stronger position than eight touched. The sequence above is chosen so that each project leaves behind infrastructure the next one needs — the surrogate becomes the RL planner’s baseline, the anticipation bundle becomes the environment strategic firms would learn in, and the spatial calibration becomes the identification backbone for everything after.

12 One direction the paper promises that is not on this list

The conclusions commit to expanding the social-learning representation “beyond a threshold model to enrich car users with social norms and behavioural control such as in the Theory of Planned Behaviour (Ajzen, 1991)”, explicitly in order to “facilitate the inclusion of policies such as information provision, media or marketing”.

This is worth adding to the portfolio for three reasons. It is the demand -side counterpart to Theme A — everything in that theme makes agents smarter about *prices*, and nothing makes them richer about *norms*. It unlocks a whole policy class (information, media, marketing) that the current model cannot represent at all, complementing the market-based instruments already characterised and the regulatory ones item 2 would add. And it bears on the same identification problem as Theme E: the binary consideration threshold $x_{i,t}$ is a very sharp behavioural assumption carrying a great deal of the model’s dynamics, and a graded alternative is both more realistic and a natural robustness check on whether the social-lock-in non-linearity of Figure S13 survives a softer specification.